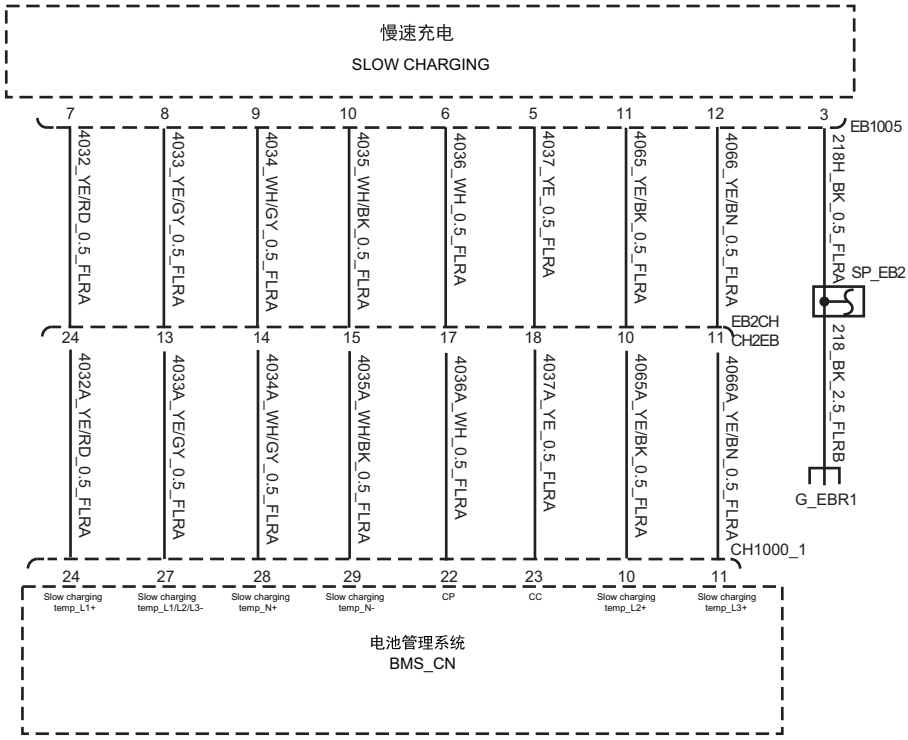
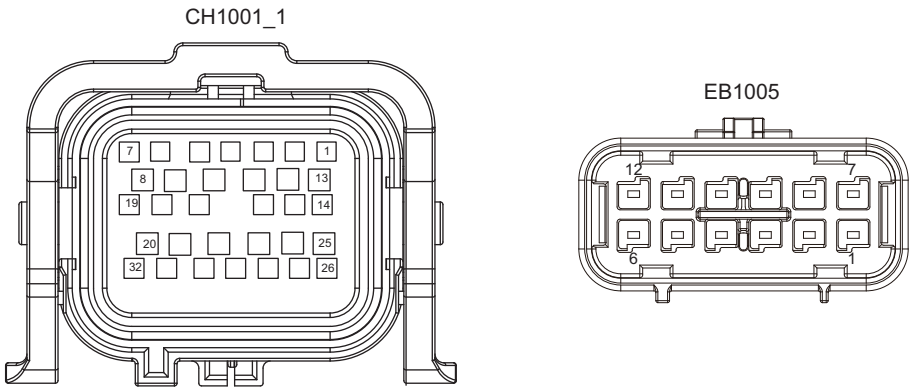
P1DF2-F1

Fault diagnosis code
P1DF2-F1：AC charging socket over temperature
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • Temperature sensor fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P1DF2-F1
AC charging socket temperature sampling value valid and temperature ⩾ [CaERRT_ACChrgPlugOverTempFltValue(1,2)=100] ℃ &&AC charging gun insertion signal is effective
To set the effect of a fault code condition
Description of circuit diagram

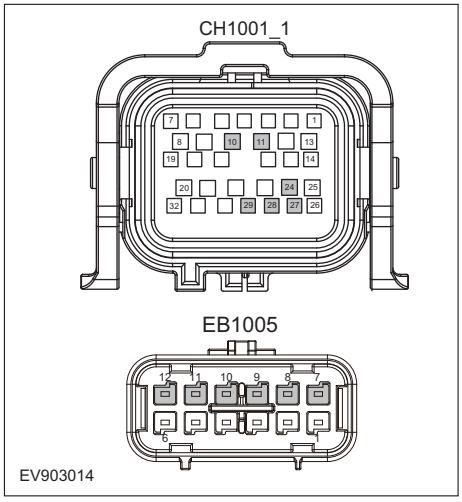
Circuit diagram

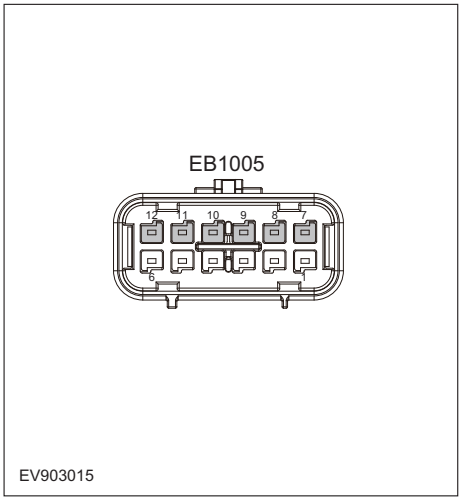


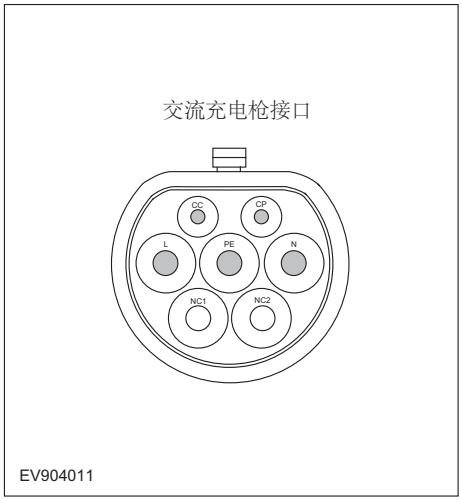


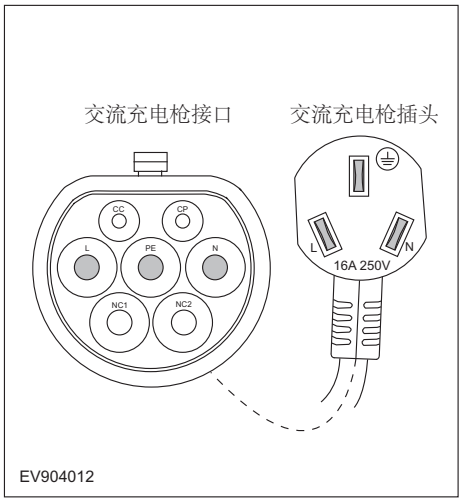
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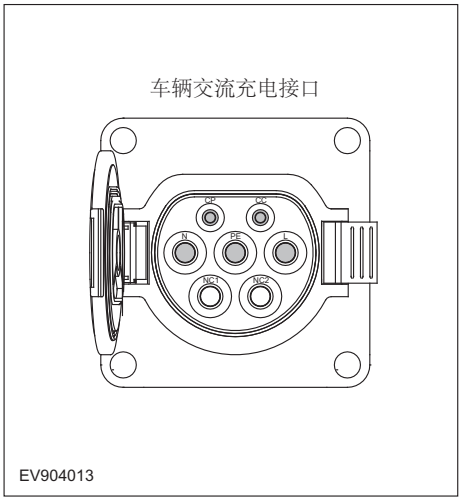
Troubleshooting steps		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check the harness plug of BMS for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the BMS with diagnostic apparatus. 4 Is there any other fault code except for P1DF2-F1?
		YES → Refer to:DTC Summary list(BMS).
NO ↓		
Step 3	Check CDU.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the CDU with diagnostic apparatus. 4 Test whether the faulted code is existed.
		YES → DTC diagnosis is performed based on specific fault codes.
NO ↓		
Step 4	Check the AC power supply.	
		1 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 2 Measure the AC power voltage. Measure circuit: voltage between the L and N terminals of the AC power socket. **Standard value: 198V~235.4V** 3 Check whether the result is normal or not?
		NO → Replace the AC power in good condition.
YES ↓		

Step 5	Check the circuit between slow charging and BMS. (Open circuit)
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug CH1000_1 of BMS. 3 Disconnect the harness plug EB1005 of slow charging. 4 Measure the resistance between the harness terminals of BMS and slow charging. Measuring circuit: resistance between terminal 10 on plug CH1000_1 and terminal 11 on plug EB1005. Measuring circuit: resistance between terminal 11 on plug CH1000_1 and terminal 12 on plug EB1005. Measuring circuit: resistance between terminal 24 on plug CH1000_1 and terminal 7 on plug EB1005. Measuring circuit: resistance between terminal 27 on plug CH1000_1 and terminal 8 on plug EB1005. Measuring circuit: resistance between terminal 28 on plug CH1000_1 and terminal 9 on plug EB1005. Measuring circuit: resistance between terminal 29 on plug CH1000_1 and terminal 10 on plug EB1005. **Standard value: < 1Ω** 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Check the circuit of slow charging. (Short circuit)
	1 Turn the ignition switch to OFF. 2 Connect the harness plug CH1000_1 of BMS. 3 Measure the resistance between the harness terminal of slow charging and ground. Measuring circuit: resistance between terminal 7 on plug EB1005 and ground. Measuring circuit: resistance between terminal 8 on plug EB1005 and ground. Measuring circuit: resistance between terminal 9 on plug EB1005 and ground. Measuring circuit: resistance between terminal 10 on plug EB1005 and ground. Measuring circuit: resistance between terminal 11 on plug EB1005 and ground. **Standard value: ∞** 4 Measure the voltage between the harness terminal of slow charging and ground. Measuring circuit: voltage between terminal 7 on plug EB1005 and ground. Measuring circuit: voltage between terminal 8 on plug EB1005 and ground. Measuring circuit: voltage between terminal 9 on plug EB1005 and ground. Measuring circuit: voltage between terminal 10 on plug EB1005 and ground. Measuring circuit: voltage between terminal 11 on plug EB1005 and ground. Measuring circuit: voltage between terminal 12 on plug EB1005 and ground. **Standard value: ≈ 0V** 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 7	Check the AC charging gun （functional check).
	1 Measure the resistance between the harness terminals of AC charging gun. Measure circuit: resistance between the L and PE terminals of the AC charging gun. **Standard value: ∞** Measure circuit: the resistance between the N and PE terminals of AC charging gun. **Standard value: ∞** Measure circuit: resistance between the CP and PE terminals of the AC charging gun. **Standard value: ∞** Measure circuit: resistance between the CC and PE terminals of the AC charging gun. **Standard value(connect the confirmation switch): ∞** **Standard value(disconnect the confirmation switch): 680 Ω** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 8	Check the AC charging gun （wiring harness conduction check).
	1 Measure the resistance between the harness terminals of AC charging gun and plug. Measure circuit: resistance between the PE terminal of the AC charging gun and grounding terminal of plug. **Standard value: ≈ 0Ω** Measure circuit: resistance between the L terminal of the AC charging gun interface and the L terminal of the plug. **Standard value: ≈ 0Ω** Measure circuit: the resistance between the N terminal of the AC charging gun interface and the N terminal of the plug. **Standard value: ≈ 0Ω** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 9	Check the AC charging port of the vehicle （functional check).
	1 Turn the ignition switch to OFF. 2 Open the vehicle AC charging port cover. 3 Measure the resistance between the harness terminals of AC charging port interface. Measure circuit: resistance between the L and PE terminals of the AC charging port. **Standard value: ∞** Measure circuit: the resistance between the N and PE terminals of AC charging port. **Standard value: ∞** Measure circuit: resistance between the CP and PE terminals of the AC charging port. **Standard value: ∞** Measure circuit: resistance between the CC and PE terminals of the AC charging port. **Standard value(connect the confirmation switch): ∞** **Standard value(disconnect the confirmation switch): 680 Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 10	Check the high-voltage cable connecting the AC charging port of the vehicle to the CDU (wiring harness conduction check).
	1 Turn the ignition switch to OFF. 2 Disconnect AC charging high-voltage cable of CDU. 3 Measure the resistance between the harness terminals of AC charging port interface and plug. Measure circuit: the resistance between the L1, L2, L3, N of AC charging port and CDU. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 11	Test whether BMS is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug CH1000_1 of BMS. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the BMS plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Replace the BMS.
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.	
Next Step ↓	
Diagnosis end.	